

Corner singularities and quasiconcavity rigidity for Robin ground states

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Abstract

We prove two results on the concavity of Robin ground states on convex polyhedra. First, for every bounded convex polygon that is neither tangential nor a rectangle, the positive parameters with a log-concave ground state form a locally finite set. An explicitly extracted real-analytic corner coefficient continues the first-variation obstruction through the finite parameter axis, disproving the general eventual log-concavity conjecture of Andrews, Clutterbuck and Hauer. Second, on every bounded convex polyhedron in dimension at least three that is not a product of circumsolids, the ground state has a nonconvex superlevel set for every sufficiently small positive parameter. This proves their polyhedral quasiconcavity conjecture. The new global rigidity argument shows that a quasiconcave solution of the first-variation Neumann problem must be quadratic: quasiconcavity removes the open-edge singularities, continuity of the gradient controls energy near the higher-codimension faces, and a Hessian-norm identity forces a constant Hessian. We also retain the discrete-exception prism consequence and an independent transverse-cone theorem under explicit angle conditions. The results are compatible with eventual concavity on smooth uniformly convex domains.

Keywords: Robin eigenfunction; log-concavity; polygonal domain; corner singularity; analytic perturbation; quasiconcavity.

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1. Introduction

For a bounded convex polygon $P \subset \mathbb{R}^2$ with nonempty interior, let u_α be the positive first Robin eigenfunction:

$$-\Delta u_\alpha = \lambda_\alpha u_\alpha \quad \text{in } P, \quad \partial_\nu u_\alpha + \alpha u_\alpha = 0 \quad \text{on } \partial P. \quad (1.1)$$

The boundary condition is understood weakly; on the interior of an edge it is classical. We fix the normalization

$$\int_P u_\alpha^2 dx = |P|. \quad (1.2)$$

In particular, $u_0 = 1$ and $\lambda_0 = 0$. Log-concavity means concavity of $\log u_\alpha$ in the open convex set P .

Andrews, Clutterbuck and Hauer [ACH] proved failure of log-concavity for sufficiently small positive Robin parameters on convex polyhedra outside a geometrically distinguished class. Their final Section 10, Conjecture 1, asks whether every fixed bounded convex domain has a threshold above which its Robin ground state is log-concave. Crasta and Fragalà [CF] established eventual log-concavity on sufficiently smooth uniformly convex domains. Ye and Zhang [YZ] recently reduced the boundary assumption to $C^{3,1}$ in every dimension. These theorems leave open the conjecture on convex domains with corners.

A polygon is *tangential* if it contains a disk tangent to every edge. This is the planar instance of a circumsolid in [ACH]. In dimension two, products of circumsolids are precisely tangential polygons and rectangles. We take polygonal descriptions with no redundant collinear vertices, so every interior angle belongs to $(0, \pi)$.

Theorem 1.1. Let P be a bounded convex polygon that is neither tangential nor a rectangle. There exist an obtuse vertex v_* and a real-analytic function $c : \mathbb{R} \rightarrow \mathbb{R}$ with

$$c(0) = 0, \quad c'(0) \neq 0, \quad (1.3)$$

such that, whenever $\alpha > 0$ and $c(\alpha) \neq 0$, neither u_α nor $\log u_\alpha$ is semiconcave on any sufficiently small neighborhood $P \cap B_r(v_*)$. Consequently

$$\{\alpha > 0 : u_\alpha \text{ is log-concave in } P\} \subseteq \{\alpha > 0 : c(\alpha) = 0\}. \quad (1.4)$$

The set on the left is locally finite, and it is empty for all sufficiently small positive parameters. In particular, for every $A > 0$ there is an $\alpha > A$ for which u_α is not log-concave.

Here semiconcavity on a convex set means that $f(x) - M|x|^2/2$ is concave for some finite constant M . The assertion near v_* refers to each sufficiently small radius, with no uniformity in α required. In fact, the exceptional zero set is finite on every bounded parameter interval, including intervals meeting zero.

Corollary 1.2. The polygon with consecutive vertices

$$(0, 0), \quad (2, 0), \quad (3, 1), \quad (1, 1) \quad (1.5)$$

does not have an eventual log-concavity threshold. There are bounded convex polyhedral counterexamples in every dimension $d \geq 2$.

The new point is the continuation of a regularity obstruction through the entire finite Robin parameter axis. The first-variation classification in [ACH] supplies the initial nonzero corner mode. We extract its continuation by a bounded functional on a fixed Sobolev space, then use the identity theorem for real-analytic functions. Thus the argument does not extrapolate a small-parameter asymptotic estimate to large parameters.

We do not determine whether log-concavity occurs at any of the exceptional parameters. Nor do we prove a classification for tangential polygons. Section 6 gives nonconvex superlevel sets on fixed prisms in every dimension at least three for all parameters outside the same discrete set. Section 8 proves the full small-parameter result in [ACH, Section 10, Conjecture 2]:

Theorem 1.3 (polyhedral quasiconcavity conjecture). On every bounded convex polyhedron $\Omega \subset \mathbb{R}^d$, $d \geq 3$, which is not a product of circumsolids, the first Robin eigenfunction has a nonconvex superlevel set for every sufficiently small positive Robin parameter.

A product of circumsolids is an orthogonal Cartesian product of polyhedra each circumscribed about a ball. Theorem 1.3 follows from a new global rigidity statement for the Neumann first variation, proved as Theorems 8.1--8.2. Its proof uses the spherical spectral lower bound and inconsistent-normal obstruction of [ACH], but does not assume their proposed higher-dimensional transverse-cone lemma. Section 7 gives an independent restricted result on that local question. The sharp Robin fundamental-gap problem remains outside this paper.

2. Analytic dependence and local regularity

We first record precisely the regularity needed for the coefficient construction. Write

$$\Gamma_\omega = \{(r \cos \theta, r \sin \theta) : r > 0, 0 < \theta < \omega\}, \quad S_R = \Gamma_\omega \cap B_R(0), \quad \beta = \frac{\pi}{\omega} > 1. \quad (2.1)$$

Lemma 2.1 (sector regularity). Suppose $z \in H^1(\Gamma_\omega)$ is supported in a fixed closed ball, has homogeneous weak Neumann data on the rays, and $\Delta z \in L^p(\Gamma_\omega)$, where $2 \leq p < \infty$. If

$$2 - \frac{2}{p} < \beta, \quad (2.2)$$

then $z \in W^{2,p}(\Gamma_\omega)$. For a fixed support radius the inclusion is continuous for the graph norm $\|z\|_{H^1} + \|\Delta z\|_{L^p}$.

Proof. This is a direct planar consequence of Dauge's wedge theorem [D, Theorem 8.1], which applies to compactly supported weak Neumann solutions on $\mathbb{R} \times \Gamma_\omega$ when $0 < 2 - 2/p < \pi/\omega$. To check the reduction, choose a nonzero $\eta \in C_c^\infty(\mathbb{R})$ and set $Z(t, x) = \eta(t)z(x)$. Then $Z \in H^1$ has compact support and

$$\Delta_{t,x} Z = \eta \Delta_x z + \eta'' z \in L^p. \quad (2.3)$$

Here $z \in L^p$ follows from the planar H^1 embedding on a bounded sector containing its support. The boundary data of Z are homogeneous Neumann. The cited theorem gives $Z \in W^{2,p}$, hence $z \in W^{2,p}$, for example by integration against a smooth function $\xi(t)$ with $\int \xi \eta = 1$.

For continuity, consider the pairs $(z, f) \in H^1 \times L^p$ with the fixed support restriction and

$$\int_{\Gamma_\omega} \nabla z \cdot \nabla \varphi = - \int_{\Gamma_\omega} f \varphi \quad \text{for all } \varphi \in C_c^\infty(\overline{\Gamma_\omega}). \quad (2.4)$$

They form a closed subspace of $H^1 \times L^p$. The map $(z, f) \mapsto z$ into $W^{2,p}$ has closed graph, since both limits agree as distributions. The closed graph theorem proves the estimate. $\square$

Lemma 2.2 (analytic eigenpair). With normalization (1.2), the maps $\alpha \mapsto \lambda_\alpha$ and $\alpha \mapsto u_\alpha$ are real analytic on $\mathbb{R}$, the latter with values in $H^1(P)$.

Proof. The Robin form

$$a_\alpha(f, g) = \int_P \nabla f \cdot \nabla g + \alpha \int_{\partial P} f g \quad (2.5)$$

has fixed domain $H^1(P)$. The trace map is bounded, and a sufficiently large scalar shift makes the form coercive locally uniformly in α . The first eigenvalue is simple and its eigenfunction can be chosen positive; see [ACH, Proposition 3.1].

For completeness, analytic dependence follows from the analytic implicit function theorem applied to

$$(\alpha, u, \lambda) \mapsto \left(a_\alpha(u, \cdot) - \lambda(u, \cdot)_{L^2(P)}, (u, u)_{L^2(P)} - |P| \right) \quad (2.6)$$

with values in $(H^1(P))^* \times \mathbb{R}$. Its derivative in (u, λ) sends (h, s) to

$$\left(B_\alpha h - s(u_\alpha, \cdot)_{L^2(P)}, 2(u_\alpha, h)_{L^2(P)} \right), \quad B_\alpha = a_\alpha - \lambda_\alpha(\cdot, \cdot)_{L^2(P)}. \quad (2.7)$$

The operator $B_\alpha : H^1 \rightarrow (H^1)^*$ is Fredholm of index zero, by a coercive shift and compactness of the L^2 embedding. Its kernel is the span of u_α , and its range is the annihilator of u_α . Pairing the first component with u_α determines s , the second component determines the component of h along u_α , and B_α is invertible on the complementary subspace. Thus (2.7) is an isomorphism. The local analytic branches remain the first eigenpair by the spectral gap. Positivity and (1.2) make them agree on overlaps, giving the asserted real-analytic maps. $\square$

At a vertex translated to the origin, let ν_1, ν_2 be the two outward unit normals. They are linearly independent. Let γ be the unique vector satisfying

$$\nu_1 \cdot \gamma = \nu_2 \cdot \gamma = -1. \quad (2.8)$$

Define the local gauge

$$w_\alpha(x) = e^{-\alpha\gamma \cdot x} u_\alpha(x). \quad (2.9)$$

It satisfies homogeneous Neumann conditions on both rays and

$$\Delta w_\alpha = F_\alpha, \quad F_\alpha = -2\alpha\gamma \cdot \nabla w_\alpha - (\lambda_\alpha + \alpha^2|\gamma|^2)w_\alpha. \quad (2.10)$$

Lemma 2.3 (regularity of the gauge). On a sufficiently small fixed sector, w_α depends real analytically on α in H^2 . Consequently F_α is real analytic in L^p for every finite p . For each fixed α , w_α is $C^{1,\delta}$ up to the vertex for some $\delta > 0$, and

$$\nabla w_\alpha(0) = 0. \quad (2.11)$$

Proof. Choose a smooth radial cutoff χ supported strictly inside the radius on which P agrees with the sector, and equal to one on a smaller sector. Extend $z_\alpha = \chi w_\alpha$ by zero to Γ_ω . Its ray normal derivative is zero. Lemma 2.2 and multiplication by the exponential show that z_α is analytic in H^1 . Moreover,

$$\Delta z_\alpha = \chi F_\alpha + 2\nabla\chi \cdot \nabla w_\alpha + (\Delta\chi)w_\alpha \quad (2.12)$$

is analytic in L^2 . The pairs $(z_\alpha, \Delta z_\alpha)$ are therefore analytic in the closed graph space in Lemma 2.1. Since $1 < \beta$, that lemma with $p = 2$ makes z_α analytic in H^2 . Planar Sobolev embedding gives analytic dependence of w_α and ∇w_α in every finite L^p on a smaller sector, and hence the claim about F_α .

Choose $p > 2$ sufficiently close to two that (2.2) holds. A second, nested radial cutoff and (2.12), now with L^p data, give $w_\alpha \in W^{2,p}$ locally. Morrey embedding gives $C^{1,\delta}$ regularity, with $0 < \delta < 1 - 2/p$. The two Neumann conditions then hold at the vertex by continuity. The independent normals force (2.11). $\square$

Lemma 2.4 (positivity at the vertex). If $\alpha > 0$, then $w_\alpha(0) > 0$.

Proof. On a sufficiently small sector put $\rho(x) = e^{2\alpha\gamma \cdot x}$ and $k = \lambda_\alpha + \alpha^2|\gamma|^2 > 0$. Equation (2.10) becomes

$$-\operatorname{div}(\rho \nabla w_\alpha) = k \rho w_\alpha. \quad (2.13)$$

The ground state is positive in the interior. It is also positive at every point in the relative interior of either edge: otherwise the Hopf boundary lemma at that smooth boundary point would contradict the Robin condition. Thus w_α has a positive minimum m on the closed outer circular arc of a small sector. Regularity from Lemma 2.3 gives continuity on its closure.

Use $\varphi = (m - w_\alpha)_+$ as a test function in (2.13), with zero trace on the outer arc and no restriction on the rays. The ray fluxes vanish, and $w_\alpha \geq 0$. Hence

$$-\int_{S_R} \rho |\nabla \varphi|^2 = k \int_{S_R} \rho w_\alpha \varphi \geq 0. \quad (2.14)$$

It follows that $\varphi = 0$, since its outer-arc trace is zero. Therefore $w_\alpha \geq m$ on the whole sector, including its vertex. $\square$

3. A bounded functional for the corner coefficient

Fix an obtuse sector, so $1 < \beta < 2$, and a radius R lying inside the region of Lemma 2.3. Set

$$\begin{aligned} a_\alpha(r) &= \frac{2}{\omega} \int_0^\omega w_\alpha(r, \theta) \cos(\beta\theta) d\theta, \\ b_\alpha(r) &= \frac{2}{\omega} \int_0^\omega F_\alpha(r, \theta) \cos(\beta\theta) d\theta. \end{aligned} \quad (3.1)$$

Lemma 3.1 (analytic coefficient). The limit

$$c(\alpha) = \lim_{r \downarrow 0} r^{-\beta} a_\alpha(r) \quad (3.2)$$

exists for every real α and is a real-analytic function of α . More precisely,

$$\begin{aligned} c(\alpha) &= R^{-\beta} a_\alpha(R) \\ &\quad - \frac{1}{2\beta} \int_0^R (s^{1-\beta} - R^{-2\beta} s^{1+\beta}) b_\alpha(s) ds. \end{aligned} \quad (3.3)$$

Proof. Angular projection of (2.10), using the homogeneous Neumann conditions, gives

$$a_\alpha'' + \frac{1}{r} a_\alpha' - \frac{\beta^2}{r^2} a_\alpha = b_\alpha. \quad (3.4)$$

This equation first holds distributionally away from zero; the functions are smooth there. Choose a finite exponent p so large that

$$p > \frac{2}{2-\beta}, \quad \kappa := 2 - \frac{2}{p} > \beta. \quad (3.5)$$

Lemma 2.3 permits this choice for F_α . Variation of constants in (3.4) gives the regular particular solution

$$A_\alpha(r) = \frac{1}{2\beta} \left[r^\beta \int_0^r s^{1-\beta} b_\alpha(s) ds - r^{-\beta} \int_0^r s^{1+\beta} b_\alpha(s) ds \right]. \quad (3.6)$$

Hölder's inequality, with planar area measure $s ds d\theta$, gives

$$|A_\alpha(r)| + r|A'_\alpha(r)| \leq Cr^\kappa \|F_\alpha\|_{L^p(S_r)}. \quad (3.7)$$

For the first integral the relevant weight is $s^{-\beta}$ relative to area measure. Its $L^{p'}$ norm is finite exactly when $\beta p' < 2$, which is equivalent to (3.5). The second integral is less singular. The homogeneous solutions of (3.4) are r^β and $r^{-\beta}$. The latter is excluded by $w_\alpha \in H^1(S_R)$, since its angular projection would have infinite H^1 energy. Thus

$$a_\alpha(r) = c(\alpha)r^\beta + A_\alpha(r). \quad (3.8)$$

Equations (3.7) and (3.5) prove (3.2). Evaluating (3.8) at R proves (3.3).

The trace functional $w \mapsto a(R)$ is bounded on H^2 on a sector extending past R . The integral in (3.3) is a bounded linear functional of $F \in L^p(S_R)$ by the same weighted Hölder estimate. Lemma 2.3 therefore proves analyticity of (3.3). The value is independent of the auxiliary radius because it equals the limit (3.2). $\square$

Two different exponents have been used for different purposes. An exponent just above two yields C^1 regularity in Lemma 2.3. A much larger exponent is allowed for the forcing F_α in Lemma 3.1. No $W^{2,p}$ regularity of w_α is asserted for that larger exponent; generally it is false when $c(\alpha) \neq 0$.

Lemma 3.2 (semiconcavity annihilates the coefficient). For $\alpha > 0$, if either u_α or $\log u_\alpha$ is semiconcave on a neighborhood of the vertex, then $c(\alpha) = 0$.

Proof. By Lemmas 2.3 and 2.4, u_α , w_α and their logarithms are C^1 on a smaller closed sector, and the positive functions are bounded away from zero there. If u_α is semiconcave, its interior Hessian has an upper bound, while

$$\Delta u_\alpha = -\lambda_\alpha u_\alpha \quad (3.9)$$

is bounded. In two dimensions a symmetric matrix with bounded trace and eigenvalues bounded above has eigenvalues bounded below as well. Therefore $D^2 u_\alpha$ is bounded in the interior of that sector. Integration along segments in the convex sector gives $u_\alpha \in C^{1,1}$ up to the vertex, and multiplication by the smooth gauge gives $w_\alpha \in C^{1,1}$.

If $\log u_\alpha$ is semiconcave, put $q = \log w_\alpha = \log u_\alpha - \alpha\gamma \cdot x$. Subtracting a linear function preserves semiconcavity. Equation (2.10) gives

$$\Delta q = -|\nabla q|^2 - 2\alpha\gamma \cdot \nabla q - (\lambda_\alpha + \alpha^2|\gamma|^2). \quad (3.10)$$

Its right hand side is bounded. The same Hessian argument gives $q \in C^{1,1}$ and hence $w_\alpha \in C^{1,1}$.

In either case, (2.11) yields

$$w_\alpha(x) = w_\alpha(0) + O(|x|^2). \quad (3.11)$$

The angular integral of $\cos(\beta\theta)$ is zero, so (3.1) gives $a_\alpha(r) = O(r^2)$. Since $\beta < 2$, (3.2) forces $c(\alpha) = 0$. The same argument works on every smaller corner neighborhood. $\square$

4. The initial nonzero mode

Let

$$v = \left. \frac{du_\alpha}{d\alpha} \right|_{\alpha=0}, \quad \mu = \frac{|\partial P|}{|P|}. \quad (4.1)$$

Differentiation of the weak eigenvalue equation and normalization gives

$$\Delta v = -\mu \quad \text{in } P, \quad \partial_\nu v = -1 \quad \text{on } \partial P, \quad \int_P v = 0. \quad (4.2)$$

We use the following established rigidity result, with regularity on the closure understood: by [ACH, Corollary 8.3], a solution of (4.2) is $C^2(\overline{P})$ if and only if P is a product of circumsolids. Thus it is not $C^2(\overline{P})$ under the hypotheses of Theorem 1.1.

Lemma 4.1 (nonzero first variation). If P is neither tangential nor a rectangle, there is an obtuse vertex for which the coefficient in Lemma 3.1 satisfies $c(0) = 0$ and $c'(0) \neq 0$.

Proof. At any vertex, translate the vertex to zero and choose γ as in (2.8). The function

$$h(x) = v(x) + \frac{\mu}{4}|x|^2 - \gamma \cdot x \quad (4.3)$$

is harmonic with homogeneous Neumann data on the two rays. Indeed, the normal derivative of $|x|^2$ vanishes on either ray. Separation of variables gives

$$h(r, \theta) = d_0 + \sum_{j \geq 1} d_j r^{j\pi/\omega} \cos\left(\frac{j\pi\theta}{\omega}\right). \quad (4.4)$$

This expansion also follows from [ACH, Proposition 5.5]. It can be obtained directly by expanding the trace on an inner arc in the Neumann cosine basis. The H^1 condition excludes negative powers and the logarithmic zero mode. On every strictly smaller sector the series and its derivatives converge away from zero. Moreover, the part with exponents strictly greater than two has second derivatives tending to zero at the vertex: square summability of the trace coefficients and the geometric factor $(r/R)^{j\pi/\omega}$ justify termwise bounds on smaller radii.

If $\omega < \pi/2$, every nonconstant exponent exceeds two. If $\omega = \pi/2$, the first term is a harmonic quadratic polynomial, and all subsequent exponents exceed two. If $\pi/2 < \omega < \pi$, only the first exponent $\beta = \pi/\omega$ lies below two, and the next is $2\beta > 2$. Therefore failure of C^2 regularity at a vertex is possible only at an obtuse vertex with $d_1 \neq 0$.

The solution v is smooth in the interior and up to the relative interiors of edges, by reflection after subtracting a local particular solution with the given normal derivative. If every vertex had a C^2 expansion, these local statements would give $v \in C^2(\overline{P})$, contrary to the rigidity result above. Fix an obtuse vertex with $d_1 \neq 0$.

At this vertex $w_0 = 1$, so $c(0) = 0$. Differentiating (2.9) and (2.10) at zero gives

$$\dot{w}_0 = v - \gamma \cdot x, \quad \dot{F}_0 = -\mu. \quad (4.5)$$

The constant forcing has zero first cosine projection. The radial quadratic term in (4.3) also has zero first cosine projection. It follows from (3.3), which may be differentiated by Lemma 3.1, and (4.4) that

$$c'(0) = d_1 \neq 0. \quad (4.6)$$

This is the claimed vertex and coefficient. $\square$

Proof of Theorem 1.1. Choose the vertex from Lemma 4.1 and its analytic coefficient from Lemma 3.1. Since $c'(0) \neq 0$, this real-analytic function is not identically zero. Its zero set is locally finite on $\mathbb{R}$, by the identity theorem, and zero is an isolated simple zero. Lemma 3.2 shows that semiconcavity of either u_α or its logarithm on any sufficiently small corner neighborhood forces $c(\alpha) = 0$. This proves the local assertion and (1.4). A locally finite subset cannot contain any interval $(A, A + 1)$, so there are arbitrarily large parameters outside the exceptional set. $\square$

5. Explicit domains and the Dirichlet limit

Proof of Corollary 1.2. The polygon (1.5) is a parallelogram with adjacent side lengths 2 and $\sqrt{2}$, and angles $\pi/4$ and $3\pi/4$. It is not a rectangle. A tangential quadrilateral has equal sums of opposite side lengths: the two tangent segments from each vertex have equal lengths, and summing them gives the identity. A tangential parallelogram must therefore have equal adjacent side lengths. Our polygon is not tangential, so Theorem 1.1 applies. At its obtuse vertices the relevant exponent is $\beta = 4/3$.

For $d > 2$, use

$$\Omega_d = P \times (0, 1)^{d-2}. \quad (5.1)$$

The positive product of the first Robin eigenfunctions on the factors satisfies the Robin condition with the same parameter on every face. Its eigenvalue is the sum of the first eigenvalues of the factors, and positivity identifies it with the ground state. Restriction to a slice with fixed interior coordinates in the interval factors is a positive constant multiple of u_α on P . Log-concavity of the product would imply log-concavity on this slice. Thus every parameter excluded for P is excluded for Ω_d . $\square$

The contrast with the Dirichlet limit concerns regularity at the corners. A nonzero coefficient in (3.2) is compatible with convergence in weaker norms and with arbitrarily small amplitude as α increases. For each fixed finite parameter with $c(\alpha) \neq 0$, however, the mode r^β with $\beta < 2$ prevents $C^{1,1}$ regularity of the gauge at the vertex. Semiconcavity would force that regularity through the bounded-trace argument of Lemma 3.2. No uniform C^2 convergence near the polygon's corners is available from a Dirichlet limit alone.

The results of [CF] and [YZ] assume smooth uniformly convex domains. Our polygonal examples satisfy neither boundary hypothesis. Approximating a polygon by smooth domains for a fixed parameter does not produce a single smooth domain with the discrete-exception property proved here, since the approximation can depend on the parameter. The remaining smooth-boundary questions must be treated separately.

The method suggests two further problems. One is to determine the exceptional parameters, or to show their absence for particular nonsymmetric polygons. Another is to understand eventual log-concavity within the tangential class, where the first-variation singular coefficient vanishes and the

present initial-mode argument gives no obstruction. These questions are distinct from the conjecture for all convex domains, which Corollary 1.2 answers negatively.

6. Nonconvex superlevel sets on fixed prisms

A positive function U on a convex domain is *quasiconcave* if every superlevel set $\{U > t\}$ is convex. For a smooth positive function, a strictly positive Hessian in a direction tangent to a level set of $\log U$ rules out quasiconcavity. We give the elementary argument in the following lemma.

Lemma 6.1 (interval lifting). Let $D \subset \mathbb{R}^m$ be an open convex set, let $f \in C^2(D)$ have bounded gradient, and suppose f is not semiconcave on D . Let I be an open interval and $h \in C^2(I)$ satisfy $h'(t_0) \neq 0$ at some $t_0 \in I$. Then

$$U(x, t) = \exp(f(x) + h(t)) \quad (6.1)$$

has a nonconvex superlevel set in $D \times I$.

Proof. If $D^2 f(x)[v, v] \leq M$ for every $x \in D$ and every unit vector v , integration along segments shows that $f - M|x|^2/2$ is concave. Therefore, failure of semiconcavity gives points $x_j \in D$ and unit vectors v_j such that

$$D^2 f(x_j)[v_j, v_j] \longrightarrow +\infty. \quad (6.2)$$

Put $F = \log U$ and

$$s_j = -\frac{\nabla f(x_j) \cdot v_j}{h'(t_0)}, \quad V_j = (v_j, s_j). \quad (6.3)$$

The numbers s_j are bounded. At $z_j = (x_j, t_0)$ we have

$$\nabla F(z_j) \cdot V_j = 0, \quad D^2 F(z_j)[V_j, V_j] = D^2 f(x_j)[v_j, v_j] + h''(t_0)s_j^2 \longrightarrow +\infty. \quad (6.4)$$

Fix j for which the last expression is positive. Since z_j is interior, Taylor's theorem gives, for sufficiently small $\varepsilon > 0$,

$$F(z_j \pm \varepsilon V_j) > F(z_j). \quad (6.5)$$

Choose b strictly between $F(z_j)$ and the smaller endpoint value. Both endpoints belong to $\{U > e^b\}$, while their midpoint does not. This proves the assertion. $\square$

Theorem 6.2 (persistent failure of quasiconcavity on prisms). Let P satisfy the hypotheses of Theorem 1.1, and let c be its analytic corner coefficient. For every integer $d \geq 3$, the first Robin eigenfunction on the fixed convex polyhedron

$$\Omega_d = P \times (0, 1)^{d-2} \quad (6.6)$$

has a nonconvex superlevel set whenever $\alpha > 0$ and $c(\alpha) \neq 0$. Consequently, the parameters at which this ground state is quasiconcave form a locally finite set, and there is no eventual quasiconcavity threshold for Ω_d .

Proof. Fix such an α . On a sufficiently small convex corner neighborhood $D = P \cap B_r(v_*)$, the function $f = \log u_\alpha$ is smooth in the interior and has bounded gradient, by Lemmas 2.3 and 2.4. It is not semiconcave there, by Theorem 1.1.

The positive first Robin eigenfunction on $(0, 1)$ can be written

$$g_\alpha(t) = \cos(k(t - 1/2)), \quad k \tan(k/2) = \alpha, \quad 0 < k < \pi. \quad (6.7)$$

The equation for k has a unique solution because its left side is strictly increasing from zero to infinity. Direct differentiation verifies the Robin condition at both endpoints; positivity identifies the eigenfunction as the ground state. For $h = \log g_\alpha$ and $t_0 = 1/4$,

$$h'(t_0) = k \tan(k/4) > 0, \quad h''(t_0) = -k^2 \sec^2(k/4). \quad (6.8)$$

Lemma 6.1 shows that $u_\alpha(x)g_\alpha(t)$ has a nonconvex superlevel set already in $D \times (0, 1)$. This product is the ground state on $P \times (0, 1)$, by the product argument in Section 5. For $d > 3$, fix the remaining interval coordinates at interior points. The resulting slice is a positive constant multiple of the same product, so the same two endpoints and midpoint witness nonconvexity. The assertions about parameters follow from the locally finite zero set of c . $\square$

In particular, the explicit parallelogram (1.5) gives a fixed prism in each dimension $d \geq 3$ with nonconvex ground-state superlevel sets for arbitrarily large Robin parameters. The small-parameter conclusion for these product domains follows already from the planar result in [ACH] and restriction to a slice. The additional conclusion here is persistence outside a discrete set over the entire positive parameter axis. No novelty is asserted for the elementary level-set criterion or separation of variables.

This extension answers the eventual-quasiconcavity question for these fixed prisms negatively. The extension is restricted to prisms; Section 8 treats the general small-parameter polyhedral conjecture. The proof uses both bounded gradient and unbounded positive Hessian near a polygonal corner; failure of log-concavity alone would not imply the interval-lifting conclusion. No single smooth uniformly convex domain with arbitrarily large bad parameters is constructed here.

7. A weighted obstruction on higher-dimensional cones

The unresolved local condition in [ACH, Lemma 9.5 and Remark 9.6] concerns the restriction of a homogeneous harmonic Neumann mode to a transverse affine slice. The following result verifies that condition for a geometric class of cones. It imposes no symmetry on the mode.

Theorem 7.1 (transverse nonconcavity). Let $\Gamma \subset \mathbb{R}^d$, $d \geq 3$, be an open full-dimensional convex polyhedral cone with outward unit face normals ν_i . Suppose a unit vector e and a number $\eta > 0$ satisfy

$$e \cdot x \geq \eta|x| \quad (x \in \bar{\Gamma}), \quad 0 < k_i := -\nu_i \cdot e < \frac{1}{\sqrt{2}} \quad \text{for every face.} \quad (7.1)$$

Let ψ be a nonzero homogeneous weak Neumann harmonic function of degree $\beta \in (1, 2)$, with $\psi \in H^1(\Gamma \cap B_R)$ for every finite R . Then ψ is not concave on the affine slice

$$K = \{x \in \Gamma : e \cdot x = 1\}. \quad (7.2)$$

Proof. We first justify the derivatives used below. A smooth radial cutoff χ preserves the zero Neumann condition on the cone faces. On a sufficiently large convex truncation $\Gamma \cap B_R$, the function $z = \chi\psi$ vanishes near the outer sphere and has

$$\Delta z = 2\nabla\chi \cdot \nabla\psi + (\Delta\chi)\psi \in L^2. \quad (7.3)$$

The standard homogeneous Neumann regularity theorem on bounded convex domains gives $z \in H^2$. The theorem and its convex approximation argument are recalled in [T, Section 4, following Theorem 4.2]. Thus ψ is locally H^2 up to the cone boundary, including its vertex. In particular, $h = \partial_e\psi$ is locally H^1 and harmonic in the interior. Even reflection gives smoothness up to every relative face interior.

Suppose that the restriction to K is concave. The first condition in (7.1) puts every nonzero cone point at positive height, so homogeneity gives concavity on every parallel slice. Put $H = D^2\psi$. Its restriction to $e^\perp$ is negative semidefinite, and harmonicity gives

$$q := H[e, e] = -\operatorname{tr}(H|_{e^\perp}) \geq 0. \quad (7.4)$$

At a relative face interior, differentiation of the Neumann condition in every face-tangent direction shows that

$$H\nu = a\nu, \quad a = H[\nu, \nu]. \quad (7.5)$$

Write $\nu = -ke + su$, where $s = \sqrt{1 - k^2}$ and $u \perp e$ is a unit vector. Taking the e and u components of (7.5), and eliminating $H[e, u]$, gives

$$(1 - k^2)H[u, u] = (1 - 2k^2)a + k^2q. \quad (7.6)$$

The left side is nonpositive, while $q \geq 0$ and $1 - 2k^2 > 0$. Hence $a \leq 0$, and therefore

$$\partial_\nu h = H[\nu, e] = -ka \geq 0 \quad (7.7)$$

on each relative face interior.

We now pass this face inequality across the polyhedral skeleton. On each bounded set, the union of strata of codimension at least two has zero H^1 capacity. Here is the needed explicit approximation. For an affine subspace of codimension two, take a cutoff equal to zero at distance at most ε^2 , equal to one at distance at least ε , and affine in the logarithm of distance between them. On a bounded set, its squared-gradient integral is $O(1/|\log \varepsilon|)$, while its difference from one tends to zero in L^2 . A finite product treats all the lower-dimensional strata, each of which lies in such a subspace. Smooth approximations can be chosen with values between zero and one.

If φ is a nonnegative smooth compactly supported test function on $\bar{\Gamma}$ whose support avoids the skeleton, harmonicity and (7.7) give

$$\int_{\Gamma} \nabla h \cdot \nabla \varphi \geq 0. \quad (7.8)$$

Multiplying an arbitrary such φ by the preceding cutoffs gives nonnegative tests converging to it in H^1 . Since h is locally H^1 , (7.8) follows for the original test as well. In particular, no unaccounted edge flux is discarded.

Use (7.8) with

$$\varphi_R(x) = e^{-e \cdot x} \chi_0(|x|/R), \quad (7.9)$$

where χ_0 is a smooth nonnegative cutoff equal to one on $[0, 1]$ and zero on $[2, \infty)$. Homogeneity and the local H^2 estimate give polynomial growth of the relevant annular Sobolev norms. The first condition in (7.1) gives exponential decay of the weight. The term containing $\nabla \chi_0(|x|/R)$ therefore tends to zero by Cauchy--Schwarz. The remaining term converges, so (7.8) yields

$$0 \leq - \int_{\Gamma} e^{-e \cdot x} \partial_e h = - \int_{\Gamma} e^{-e \cdot x} q \leq 0. \quad (7.10)$$

It follows that $q = 0$ in the interior. The negative-semidefinite transverse Hessian block then has zero trace and must vanish. Writing $x = te + z$, $z \perp e$, connectedness of each transverse slice and homogeneity give constants $A \in \mathbb{R}$ and $b \in e^\perp$ such that

$$\psi(te + z) = At^\beta + t^{\beta-1} b \cdot z. \quad (7.11)$$

Harmonicity becomes

$$0 = (\beta - 1)t^{\beta-3}(\beta At + (\beta - 2)b \cdot z). \quad (7.12)$$

Since $1 < \beta < 2$ and Γ has nonempty interior, $A = 0$ and $b = 0$. This contradicts the nonzero mode and proves the theorem. $\square$

7.1. Consequence for the small-parameter Robin question

For a consistent-normal cone, let γ satisfy $\gamma \cdot \nu_i = -1$ for every face and set $e = \gamma/|\gamma|$. The second condition in (7.1) becomes $|\gamma| > \sqrt{2}$; the first condition is a separate strict-dual requirement.

Corollary 7.2. Let $\Omega \subset \mathbb{R}^d$, $d \geq 3$, be a bounded convex polyhedron. Suppose the first Neumann variation v of its normalized Robin ground state has at a boundary point x_0 a nonzero leading nonlinear homogeneous Neumann mode ψ of degree $\beta \in (1, 2)$. Suppose also that the tangent cone has consistent normals and satisfies (7.1) with $e = \gamma/|\gamma|$. Then the Robin ground state on Ω has a nonconvex superlevel set for every sufficiently small positive Robin parameter.

Proof. The expansion and interior derivative estimates in [ACH, Section 9] give, on each fixed compact subset of the tangent cone,

$$\begin{aligned} \nabla v(x_0 + \rho x) &= \gamma + \rho^{\beta-1} \nabla \psi(x) + o(\rho^{\beta-1}), \\ D^2 v(x_0 + \rho x) &= \rho^{\beta-2} D^2 \psi(x) + o(\rho^{\beta-2}). \end{aligned} \quad (7.13)$$

The lower-order quadratic particular solution is included in the remainders. By Theorem 7.1 there are an interior point x and a vector $\xi \perp e$ with $D^2 \psi(x)[\xi, \xi] > 0$. Correct ξ to $\xi + a_\rho e$ so that it is orthogonal to $\nabla v(x_0 + \rho x)$. Since $\gamma \cdot e = |\gamma| > 0$, the first line of (7.13) gives $a_\rho = O(\rho^{\beta-1})$. The second line gives strictly positive second derivative in this corrected direction for small ρ . Taylor's theorem at one fixed such interior point produces two endpoints whose v values strictly exceed that of their midpoint.

The Neumann perturbation $u_\alpha = 1 + \alpha v + o(\alpha)$ from [ACH, Proposition 3.1] holds uniformly on these three fixed points. Thus the strict midpoint inequality persists for u_α for all sufficiently small $\alpha > 0$. This is the transfer mechanism of [ACH, Lemma 9.5]. $\square$

The leading nonzero mode in this corollary is a hypothesis. No assertion is made that a chosen vertex necessarily has a nonzero coefficient, or that every polyhedron has a vertex satisfying (7.1).

7.2. Nonempty scope and the remaining angle range

Theorem 7.1 is not vacuous in its degree interval. Fix $0 < k < 1/\sqrt{2}$ and put $s = \sqrt{1 - k^2}$. In coordinates (x, y, t) , consider the cone with three outward normals

$$\begin{aligned}\nu_1 &= (s, 0, -k), \\ \nu_2 &= (-s \cos \varepsilon, s \sin \varepsilon, -k), \\ \nu_3 &= (-s \cos(2\varepsilon), -s \sin(2\varepsilon), -k),\end{aligned}\quad 0 < \varepsilon < \pi/6. \quad (7.14)$$

The transverse normals positively span $\mathbb{R}^2$, so the section $t = 1$ is a bounded triangle and t is strictly positive on every nonzero direction in the closed cone. Thus (7.1) holds with $e = (0, 0, 1)$ and a positive η depending on ε .

As $\varepsilon \downarrow 0$, the spherical sections converge almost everywhere to the spherical section of $\{|x| < (k/s)t\} \times \mathbb{R}_y$. On this limiting section, y has mean zero and Neumann eigenvalue two. Indeed, latitude integration gives $\int y^2 = |A_0|/3$, while $|\nabla_{S^2} y|^2 = 1 - y^2$. The Rayleigh quotient on the finite sections, tested with y minus its mean, therefore tends to two by bounded convergence. On every finite pointed cone, [ACH, Theorem 9.1] gives $\lambda_1 > 2$, since equality requires a linear factor. Hence, for sufficiently small positive ε ,

$$2 < \lambda_1 < 6, \quad 1 < \beta_1 < 2, \quad \beta_1(\beta_1 + 1) = \lambda_1. \quad (7.15)$$

No explicit cutoff in ε or eigenfunction parity is needed.

The angle condition cannot simply be deleted from the face-sign step. For example, with $e = (1, 0, 0)$,

$$\nu = (-\sqrt{3}/2, 1/2, 0), \quad H = \begin{pmatrix} 1 & -\sqrt{3} & 0 \\ -\sqrt{3} & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (7.16)$$

we have $\text{tr } H = 0$, $H|_{e^\perp} \preceq 0$, and $H\nu = 2\nu$. The normal second derivative is positive and the desired face-flux sign reverses. This matrix is an obstruction to that algebraic proof step, not a counterexample to the cone conjecture. The borderline angle, the larger-angle range, and cones without the strict-dual property require further arguments. Section 8 resolves the general polyhedral conjecture by a different, global argument.

8. Quasiconcavity forces quadraticity on every convex polyhedron

We now remove the geometric restrictions from the small-parameter result. A *product of circumsolids* has the meaning in [ACH]: after a rigid motion, the domain is an orthogonal Cartesian product of convex polyhedra, each of which contains a ball tangent to all its facets. Intervals are allowed as factors.

Theorem 8.1. Let $\Omega \subset \mathbb{R}^d$, $d \geq 3$, be a bounded convex polyhedron with nonempty interior. If Ω is not a product of circumsolids, then there exists $\alpha_0 > 0$ such that its first Robin eigenfunction has a nonconvex superlevel set for every $0 < \alpha < \alpha_0$.

This is [ACH, Section 10, Conjecture 2]. Its proof uses global rigidity rather than an unrestricted version of the transverse cone theorem in Section 7. The analytic statement behind it is the following.

Theorem 8.2. Put $\mu = |\partial\Omega|/|\Omega|$, and let $v \in H^1(\Omega)$ solve

$$\Delta v = -\mu \quad \text{in } \Omega, \quad \partial_\nu v = -1 \quad \text{on } \partial\Omega. \quad (8.1)$$

If v is quasiconcave, then v is a quadratic polynomial.

Here quasiconcavity means that every superlevel set in the open domain is convex. The additive constant in v is immaterial. We first establish three regularity and energy facts, keeping their hypotheses separate.

8.1. Continuous gradients from compatible face data

At $z \in \partial\Omega$, call the incident normals *consistent* if there is a vector γ_z with $\gamma_z \cdot \nu_i = -1$ on every incident facet. We use two standard analytic inputs. On a bounded convex domain, a weak homogeneous-Neumann solution with Laplacian in L^2 belongs to H^2 and satisfies

$$\|D^2 f\|_{L^2} \leq \|\Delta f\|_{L^2}. \quad (8.2)$$

The scalar approximation argument is recalled in [T, Section 4]. Maz'ya's gradient estimate [M] also gives $\|\nabla f\|_\infty \leq C\|\Delta f\|_{L^q}$ for each fixed bounded convex domain and $q > d$; the additive mean of f does not affect this estimate. Both facts apply on convex cones truncated by balls. Radial cutoffs have zero normal derivative on the planar cone faces and can be chosen to vanish near the spherical boundary.

Lemma 8.3. If all boundary points have consistent normals, the solution of (8.1) belongs to $H^2(\Omega)$ and its gradient extends continuously to $\overline{\Omega}$.

Proof. Fix z and a conic neighborhood of radius R_z . In coordinates centered at z , the function

$$h(x) = v(z + x) + \frac{\mu}{2d}|x|^2 - \gamma_z \cdot x \quad (8.3)$$

is harmonic with homogeneous Neumann data. Applying (8.2) to a radial cutoff of h proves local H^2 regularity. The cutoff forcing is in L^2 because $h \in H^1$. Finitely many such neighborhoods, together with interior neighborhoods, prove the global assertion.

We spell out why the homogeneous expansion also controls the gradient uniformly up to the cone faces. For a fixed cone Γ , set $A = \Gamma \cap \mathbb{S}^{d-1}$ and take an orthonormal Neumann eigenbasis Φ_j on A . Write

$$\lambda_j = \beta_j(\beta_j + d - 2), \quad \psi_j(r\theta) = r^{\beta_j} \Phi_j(\theta). \quad (8.4)$$

The cone expansion of a weak harmonic Neumann function contains only these nonnegative-degree terms; see [ACH, arXiv v2, Proposition 4.4]. By [ACH, Theorem 9.1], every positive degree is at least one, and every degree-one mode is linear. Discreteness gives a positive gap between one and the next degree.

For clarity, the spectral bounds used below need only have some polynomial exponent. The Sobolev inequality on the fixed Lipschitz spherical domain, tested in the eigenfunction equation with $|\Phi|^{p-2}\Phi$, gives

$$\|\Phi\|_{p\kappa} \leq [Cp^2(1+\lambda)]^{1/p} \|\Phi\|_p \quad (p \geq 2) \quad (8.5)$$

for some fixed $\kappa > 1$. Iterating $p = 2\kappa^m$ proves an L^∞ bound polynomial in $1 + \lambda$. There is also a polynomial counting bound for the eigenvalues. One way to see this without a sharp Weyl law is to use an H^1 extension in finitely many spherical coordinate charts. The resulting map into a finite sum of cube spaces is injective, has an L^2 lower bound, and is bounded in H^1 . Its restriction to the spectral subspace with eigenvalues at most Λ has Rayleigh quotient at most $C(1 + \Lambda)$. The min-max principle and the explicit Neumann spectra of cubes bound the dimension of that subspace by a polynomial in $1 + \Lambda$.

Apply [M] to a radial cutoff of ψ_j on $\Gamma \cap B_2$, equal to one on B_1 and zero near the outer sphere. Since $\partial_r \psi_j = \beta_j r^{\beta_j-1} \Phi_j$, its Laplacian involves no angular derivative of Φ_j . For a sufficiently large fixed exponent N , the preceding bounds imply

$$\|\Phi_j\|_\infty \leq C(1 + \beta_j)^N, \quad \|\nabla \psi_j\|_{L^\infty(\Gamma \cap B_1)} \leq C(1 + \beta_j)^N 2^{\beta_j}. \quad (8.6)$$

Choose an almost-everywhere reference radius at which the spherical trace of h is in L^2 , and rescale it to one. The expansion coefficients then satisfy $\sum_j |a_j|^2 < \infty$. Remove the constant and linear terms and put $\delta = \min_{\beta_j > 1} (\beta_j - 1) > 0$. Homogeneity, (8.6), and Cauchy--Schwarz give, for $0 < r \leq 1/8$,

$$\sup_{\Gamma \cap B_r} \left| \nabla \sum_{\beta_j > 1} a_j \psi_j \right| \leq C r^\delta \left(\sum_j |a_j|^2 \right)^{1/2}. \quad (8.7)$$

Indeed, after factoring r^δ from each term, the remaining squared series is bounded by a polynomially weighted geometric series in β_j , with ratio controlled by $2r \leq 1/4$. The counting bound makes it summable uniformly in r . The same estimate proves convergence of the derivative series in L^∞ ; its sum agrees with the weak derivative and with the classical interior derivative.

Thus the interior gradient has a limit at every z , equal to the linear coefficient in (8.3) plus γ_z . Interior regularity and this limit define a continuous gradient on the closure. Continuity along the boundary follows as well: approximate a boundary point approaching z by an interior point where the gradient is arbitrarily close to that boundary limit. Compactness now gives a modulus

$$\omega(t) = \sup_{\substack{x, y \in \bar{\Omega} \\ |x-y| \leq t}} |\nabla v(x) - \nabla v(y)| \longrightarrow 0 \quad (t \downarrow 0). \quad (8.8)$$

Finally, continuity from a relative facet interior shows that $\nabla v(z) \cdot \nu_i = -1$ on every incident facet. No uniform corner exponent is needed. $\square$

8.2. Quasiconcavity removes the open-edge singularities

Let S be the union of the closed faces of codimension at least three. It has dimension at most $d - 3$.

Lemma 8.4. If v is quasiconcave, all its boundary normals are consistent, and D^2v is locally bounded on $\bar{\Omega} \setminus S$.

Proof. The inconsistent-normal argument in [ACH, Section 9, inconsistent-normal case] produces a nonconvex superlevel set of v whenever a boundary point has inconsistent normals. Hence Lemma 8.3 applies.

At a relative interior point of a codimension-two face, the tangent cone is $W \times \mathbb{R}^{d-2}$, where W is a planar wedge of angle $\vartheta \in (0, \pi)$. We first classify its homogeneous harmonic Neumann modes of degree $\beta \in (1, 2]$. For a direction t in the linear factor, $\partial_t \psi$ is a weak harmonic Neumann function, by tangential difference quotients. It is locally H^1 by (8.2), and is homogeneous of degree $\beta - 1$.

If $1 < \beta < 2$, the spectral lower bound [ACH, Theorem 9.1] forces this derivative to vanish. The mode is therefore planar. The only possible degree in this range is $\beta = \pi/\vartheta$, with $\vartheta > \pi/2$, and in bisector coordinates it is a nonzero multiple of

$$\psi(r \cos \theta, r \sin \theta, t) = r^\beta \sin(\beta \theta), \quad -\vartheta/2 < \theta < \vartheta/2. \quad (8.9)$$

On a line perpendicular to the inward bisector this is odd and nonaffine, so neither sign is concave. To check nonaffinity, affinity would, by oddness and homogeneity, give $cx^{\beta-1}y$ in bisector coordinates. For $c \neq 0$ this is harmonic only when $\beta = 1$ or 2 .

The normal-plane projection of $\gamma = \nabla v(z)$ is the inward bisector of length $1/\sin(\vartheta/2)$. Its possible component along the edge does not affect a normal-plane direction ξ perpendicular to that bisector. Thus (8.9), with either sign, gives an interior cone point x and such a ξ with $D^2\psi(x)[\xi, \xi] > 0$. We may choose x with zero edge component and then scale it so that $\gamma \cdot x > 0$ is fixed.

If a nonzero mode (8.9) occurred in the expansion of v at z , it would be the leading nonlinear mode below degree two. At $z + rx$, the expansion gives $\nabla v = \gamma + o(1)$ and

$$D^2v = r^{\beta-2}(D^2\psi(x) + o(1)), \quad (8.10)$$

with its nonzero coefficient absorbed into ψ . Replace ξ by $\xi + c(r)x$, choosing $c(r) = o(1)$ so that it is exactly perpendicular to $\nabla v(z + rx)$. Its Hessian value is positive for small r . The two nearby points in opposite directions then have values strictly larger than the midpoint, contradicting quasiconcavity. This is the tangent correction of [ACH, Lemma 9.5]. Consequently every degree in $(1, 2)$ is absent at every open edge point.

We give the local bounded-Hessian argument explicitly. Degree-two modes on $W \times \mathbb{R}^{d-2}$ are quadratic. Indeed, their derivatives in the linear-factor directions have degree one and hence are linear Neumann functions, whose gradients lie in that linear factor. Integrating gives

$$\psi(y, t) = \tfrac{1}{2}t^T B t + \phi(y), \quad \Delta_y \phi = -\operatorname{tr} B. \quad (8.11)$$

Here B is symmetric and ϕ is homogeneous of degree two with Neumann ray data. After adding $(\operatorname{tr} B)|y|^2/4$, it is a planar harmonic mode of degree two. Such a mode is either zero or, when $\vartheta = \pi/2$, a harmonic quadratic. Thus (8.11) is quadratic in all cases.

On a compact portion of an open edge, choose one fixed conic radius. The spherical L^2 norms of the local harmonic functions (8.3) are uniformly bounded, since v and its gradient are continuous on the closure. The spectral L^∞ bound in (8.6), the counting bound, absence of degrees in $(1, 2)$, and the quadraticity just proved show that at every edge center z there is a quadratic polynomial Q_z , with uniformly bounded coefficients, such that

$$|v(z+x) - Q_z(x)| \leq C|x|^{2+\delta_2} \quad (8.12)$$

for a fixed $\delta_2 > 0$. Here the polynomial includes the particular solution in (8.3); the remainder is harmonic with homogeneous Neumann data on the two faces. The estimate follows by summing the degrees strictly greater than two, exactly as for (8.7), now without differentiating. The first such degree has a positive gap above two.

For an interior point at distance r from the edge, project to an edge center z . A ball of radius $c(\vartheta)r$ about the point meets at most one of the two faces. If it meets a face, even reflection of the harmonic remainder across that face gives the interior harmonic estimate. Using (8.12) on the original and reflected points yields

$$|D^2v - D^2Q_z| \leq Cr^{\delta_2}. \quad (8.13)$$

This proves local boundedness at the open edges. Relative facet interiors are smooth by even reflection after subtracting the normal affine term. These neighborhoods cover $\bar{\Omega} \setminus S$. In particular, no higher-dimensional regularity criterion for unspecified degree-two modes is being assumed. $\square$

8.3. Energy near the higher-codimension faces

Lemma 8.5. Under the hypotheses of Lemma 8.3, with $H = D^2v$, there are constants C and $\varepsilon_0 > 0$ such that

$$\int_{\{\text{dist}(x,S) < \varepsilon\}} |H|^2 dx \leq C(\varepsilon^3 + \varepsilon \omega(C\varepsilon)^2) \quad (0 < \varepsilon < \varepsilon_0). \quad (8.14)$$

All integrals in this subsection are over Ω . In particular the left side is $o(\varepsilon)$.

Proof. A finite union of bounded faces of dimension at most $d-3$ has an ε -net of at most $C\varepsilon^{-(d-3)}$ points, and its ε -tube has volume at most $C\varepsilon^3$. We need a uniform conic patch near each net point; a fixed cone radius at all points of S would not be valid.

Write the polytope by unit facet inequalities $\nu_j \cdot x \leq b_j$, $1 \leq j \leq m$, and put $s_j(x) = b_j - \nu_j \cdot x$. For every feasible nonempty subset J , let F_J be the nonempty face where all its inequalities are equalities. There is a constant C_0 , independent of J , such that

$$\text{dist}(z, F_J) \leq C_0 \max_{j \in J} s_j(z), \quad z \in \bar{\Omega}. \quad (8.15)$$

To prove it, write z as a convex combination of the finitely many vertices. Each vertex outside F_J has a positive total J -slack. The minimum of these positive totals, over the finitely many choices, is positive. Thus the total weight of vertices outside F_J is bounded by a constant times $\max_{j \in J} s_j(z)$. Move that weight to any vertex of F_J and use the diameter bound. For infeasible J , compactness instead gives $\min_z \max_{j \in J} s_j(z) > 0$.

Fix $z \in S$ and a small ε . Set $A_0 = 4$ and $A_{\ell+1} = 10(C_0 + 1)A_\ell$. Among $m+2$ consecutive sets

$$J_\ell = \{j : s_j(z) \leq A_\ell \varepsilon\} \quad (8.16)$$

two adjacent sets coincide. Call the common set J . For uniformly small ε , it is feasible by the infeasible-set lower bounds. Choose $w \in F_J$ with $|w - z| \leq C_0 A_\ell \varepsilon$. The set J includes the facets

incident to z , so $w \in S$. Put $R = 2(C_0 + 1)A_\ell\varepsilon$. Then $B_{2\varepsilon}(z) \subset B_R(w)$. For $j \notin J$, its supporting plane is farther than

$$[10(C_0 + 1) - C_0]A_\ell\varepsilon > 2R \quad (8.17)$$

from w . Hence inside $B_{2R}(w)$ the domain is exactly its tangent cone at w . There are only finitely many threshold levels, so R is comparable to ε with constants depending only on the fixed polytope.

Subtract the affine tangent at w , writing $g(x) = v(x) - v(w) - \nabla v(w) \cdot (x - w)$. It has homogeneous Neumann data on the incident facets, by Lemma 8.3. On this conic patch, convexity and (8.8) imply

$$|\nabla g| \leq \omega(C\varepsilon), \quad |g| \leq CR\omega(C\varepsilon). \quad (8.18)$$

Take a radial cutoff ρ equal to one on $B_R(w)$ and supported strictly inside $B_{2R}(w)$, with derivatives bounded by C/R and C/R^2 . Apply (8.2) to ρg on the cone truncated at radius $2R$. Since

$$\Delta(\rho g) = -\mu\rho + 2\nabla\rho \cdot \nabla g + g\Delta\rho, \quad (8.19)$$

we obtain

$$\int_{\Omega \cap B_R(w)} |H|^2 \leq C(R^d + R^{d-2}\omega(C\varepsilon)^2). \quad (8.20)$$

The balls $B_{2\varepsilon}(z)$ for the net points cover the ε -tube of S . Sum (8.20) over at most $C\varepsilon^{-(d-3)}$ balls to get (8.14). Overlaps only enlarge this upper bound. $\square$

8.4. A Hessian-norm identity and the proof of rigidity

Proof of Theorem 8.2. Assume v is quasiconcave. Lemmas 8.3–8.5 apply. In the interior, every component of $H = D^2v$ is harmonic, since $\Delta v = -\mu$ is constant. At a relative facet interior, subtract an affine function with normal derivative -1 and reflect the result evenly. The Hessian transforms by orthogonal conjugation. Consequently, on that facet,

$$H : \partial_\nu H = 0, \quad \partial_\nu |H| = 0. \quad (8.21)$$

In coordinates with the normal as the last axis, the mixed normal-tangential entries of H vanish, while the normal derivatives of the other entries vanish. This also verifies (8.21) directly.

Since $\text{tr } H = -\mu \neq 0$, the Frobenius norm $q = |H|$ satisfies $q \geq \mu/\sqrt{d} > 0$. Its interior Laplacian is

$$D := \Delta q = \sum_{k=1}^d \left(\frac{|\partial_k H|^2}{q} - \frac{(H : \partial_k H)^2}{q^3} \right) \geq 0. \quad (8.22)$$

For any Lipschitz cutoff χ vanishing on a neighborhood of S , integration by parts gives

$$\int_{\Omega} \chi^2 |\nabla H|^2 \leq 4 \int_{\Omega} |\nabla \chi|^2 |H|^2, \quad (8.23)$$

and

$$\int_{\Omega} \chi^2 D = -2 \int_{\Omega} \chi \nabla \chi \cdot \nabla q. \quad (8.24)$$

Here is the justification at the open edges. On each compact set away from S , Lemma 8.4 bounds H . Around a codimension-two affine face, a cutoff increasing logarithmically from zero at distance δ^2 to one at distance δ has squared gradient integral $O(1/|\log \delta|)$. Products of finitely many such cutoffs remove the edges. First integrate the harmonic equations for H against $\chi^2 H$ times the square of this auxiliary cutoff. The facet boundary terms vanish by (8.21). Cauchy--Schwarz gives (8.23) with the auxiliary cutoff; boundedness of H makes its cutoff-energy error tend to zero. Fatou first gives local L^2 control of ∇H up to those edges. One can then pass in the identity itself: all terms containing the auxiliary cutoff gradient tend to zero by Cauchy--Schwarz and its vanishing L^2 norm. Applying the same argument to $\Delta q = D$, with $|\nabla q| \leq |\nabla H|$, proves (8.24). The integrand D is locally integrable there since q is bounded below and $D \leq |\nabla H|^2/q$. Thus no edge measure or uncomputed boundary term is discarded.

Choose χ_ε to be zero when $\text{dist}(x, S) \leq \varepsilon$ and one when that distance is at least 2ε , with $|\nabla \chi_\varepsilon| \leq C/\varepsilon$. Define

$$A_\varepsilon = \int_\Omega |\nabla \chi_\varepsilon|^2 |H|^2, \quad B_\varepsilon = \int_\Omega |\nabla \chi_\varepsilon|^2. \quad (8.25)$$

The tube-volume bound and Lemma 8.5 imply

$$A_\varepsilon \leq C \left(\varepsilon + \frac{\omega(C\varepsilon)^2}{\varepsilon} \right), \quad B_\varepsilon \leq C\varepsilon. \quad (8.26)$$

Although A_ε need not tend to zero, its product with B_ε does. Combining (8.23)--(8.26) yields

$$0 \leq \int_\Omega \chi_\varepsilon^2 D \leq 4\sqrt{A_\varepsilon B_\varepsilon} \leq C(\varepsilon + \omega(C\varepsilon)) \longrightarrow 0. \quad (8.27)$$

Fatou's lemma gives $D = 0$ in the interior. Each summand in (8.22) is nonnegative, so equality in Cauchy--Schwarz gives $\partial_k H = a_k H$ pointwise for some scalar a_k . Taking traces gives $0 = -\mu a_k$, hence $a_k = 0$. Thus H is constant, and v is quadratic on the connected domain. This argument neither assumes global H^3 regularity nor infers bounded Hessians from continuity of the gradient. $\square$

Proof of Theorem 8.1. The quadratic classification [ACH, Corollaries 8.2--8.3] says that a quadratic solution of (8.1) can exist only on a product of circumsolids. Thus Theorem 8.2 implies that v is not quasiconcave when Ω is not such a product. There are interior points x, y , their midpoint m , and $a, \tau \in \mathbb{R}$ with $\tau > 0$ such that

$$v(x), v(y) > a + \tau, \quad v(m) < a - \tau. \quad (8.28)$$

Indeed, continuity makes midpoint quasiconcavity equivalent to quasiconcavity, by dyadic subdivision. The normalized first eigenfunction has the perturbation expansion [ACH, Proposition 3.1]

$$u_\alpha = 1 + \alpha v + o(\alpha) \quad \text{uniformly on } \overline{\Omega}. \quad (8.29)$$

For all sufficiently small positive α , (8.28)--(8.29) put x, y in $\{u_\alpha > 1 + \alpha a\}$ and exclude m . This proves the asserted nonconvexity for the entire interval $(0, \alpha_0)$. $\square$

The recent work of Edelen and Li [EL] proves sharp Neumann spectral and regularity results under non-obtuse dihedral-angle hypotheses. Their Lemma 2.5 uses logarithmic cutoffs to justify a tensor energy identity when the tensor is bounded. The open-edge cutoff step above is of the same standard

type. The additional estimate (8.14) and the use of the norm rather than its square allow the higher-codimension part to remain potentially unbounded, while quasiconcavity supplies the needed edge regularity even at obtuse edges. We do not use the spectral-gap theorem of [EL].

The unrestricted transverse-cone assertion suggested by [ACH, Remark 9.6] remains a separate local question. Theorem 8.1 resolves their polyhedral quasiconcavity conjecture through the global Neumann problem. It gives no assertion about large parameters on every polyhedron, or about the sharp Robin fundamental gap.

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